

**University Interscholastic League**  
**2024 – 2025 Elementary Number Sense Test C**

Contestant's Number \_\_\_\_\_

|                 |              |                 |
|-----------------|--------------|-----------------|
| Final           |              |                 |
| 2 <sup>nd</sup> |              |                 |
| 1 <sup>st</sup> |              |                 |
|                 | <b>Score</b> | <b>Initials</b> |

**Read Directions Carefully  
Before Beginning Test**

**Do Not Unfold This Sheet  
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (\*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.  
**Stop – Wait for Signal!**

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| <p>(1) <math>71 + 26 =</math> _____</p> <p>(2) <math>38 + 83 =</math> _____</p> <p>(3) <math>701 - 46 =</math> _____</p> <p>(4) <math>204 \div 4 =</math> _____</p> <p>(5) <math>6 \times 23 =</math> _____</p> <p>(6) <math>472 - 363 =</math> _____</p> <p>(7) <math>32 + 14 + 18 =</math> _____</p> <p>(8) <math>49 \times 11 =</math> _____</p> <p>(9) <math>50 \times 76 =</math> _____</p> <p>*(10) <math>5 + 792 + 1993 =</math> _____</p> <p>(11) <math>36 \times 25 =</math> _____</p> <p>(12) 38670.4782 rounded to the thousands place is<br/>           _____</p> <p>(13) Which digit is in the thousands place in<br/>           743914.20685? _____</p> <p>(14) <math>51 \times 49 =</math> _____</p> <p>(15) What is the remainder for <math>64388 \div 9</math>? _____</p> <p>(16) The number of odd whole numbers between 17 and<br/>           39 is _____</p> <p>(17) <math>2 \times 10^{-3} + 3 \times 10^2 + 8 \times 10^{-2} =</math> _____ (decimal)</p> <p>(18) <math>24 \times 35 - 24 \times 5 =</math> _____</p> <p>(19) XLIV = _____ (Arabic Numeral)</p> | <p>*(20) <math>42 \times 667 =</math> _____</p> <p>(21) <math>18 + 12 \div 3 \times 2 =</math> _____</p> <p>(22) <math>26 + 22 + 18 + 14 =</math> _____</p> <p>(23) <math>\frac{7}{12}</math> hour = _____ minutes</p> <p>(24) <math>286 \times 7 =</math> _____</p> <p>(25) <math>\frac{13}{24} - \frac{3}{24} =</math> _____</p> <p>(26) <math>103 \times 102 =</math> _____</p> <p>(27) 84% = _____ common fraction</p> <p>(28) If 12 ♥ costs 27¢, then 16 ♥ cost _____ ¢</p> <p>(29) <math>24 \div \frac{3}{10} =</math> _____</p> <p>*(30) <math>639 \times 126 + 86 =</math> _____</p> <p>(31) \$27.75 = _____ quarters</p> <p>(32) The smallest prime number between 50 and 20 is<br/>           _____</p> <p>(33) 48 ounces = _____ pints</p> <p>(34) <math>44 \times 21 =</math> _____</p> <p>(35) <math>62\frac{1}{2}\%</math> = _____ common fraction</p> <p>(36) The LCM of 36 and 24 is _____</p> <p>(37) <math>45 \times 15 =</math> _____</p> |
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- (38) If 12 apples cost \$2.40, then 8 apples cost \$\_\_\_\_\_
- (39) 12 is to 20 as 5 is to \_\_\_\_\_
- \*(40)  $318 \times 625 =$  \_\_\_\_\_
- (41)  $\frac{11}{24} + \frac{7}{24} =$  \_\_\_\_\_ (common fraction)
- (42)  $15^2 - 5 =$  \_\_\_\_\_
- (43) If  $f(x) = 2x - 7$ , then  $f(12) =$  \_\_\_\_\_
- (44) The diameter of a circle with an area of  $100\pi$  is  
\_\_\_\_\_
- (45)  $5^3 \div 5^0 =$  \_\_\_\_\_
- (46)  $8\frac{1}{8} \times 8\frac{7}{8} =$  \_\_\_\_\_ (mixed number)
- (47) What is the perimeter of a square with area 49? \_\_\_\_\_
- (48)  $55\frac{5}{9} \% \times 36 =$  \_\_\_\_\_
- (49) If there are 112 grapes to a pound, then 1 ounce equals how many grapes? \_\_\_\_\_
- \*(50)  $299\frac{1}{70} \times 402 =$  \_\_\_\_\_
- (51) What is the number, **k**, in the sequence:  
-1, 2, 7, **k**, 23, 34, ...? \_\_\_\_\_
- (52)  $5\frac{7}{8} + 7\frac{3}{4} =$  \_\_\_\_\_ (mixed number)
- (53)  $\sqrt[3]{27} =$  \_\_\_\_\_
- (54) A square with area 49 is located inside a square with perimeter 52. What is the area between the squares?  
\_\_\_\_\_
- (55) 34 (base 6) = \_\_\_\_\_ (base 10)
- (56) The number of elements in  $\{1, 2, 3, 6\} \cap \{1, 2, 3, 4\}$  is \_\_\_\_\_
- (57) What is the perimeter of a regular octagon with side  $6\frac{1}{4}$ ? \_\_\_\_\_
- (58)  $75 \times 28 =$  \_\_\_\_\_
- (59)  $22\frac{7}{9} - 4\frac{1}{3} \times 4\frac{2}{3} =$  \_\_\_\_\_
- \*(60) 250 inches = \_\_\_\_\_ centimeters
- (61) 32 (base 4) = \_\_\_\_\_ (base 2)
- (62)  $(-125) \times (-4) =$  \_\_\_\_\_
- (63) What is the probability a getting a sum of 5 when rolling a pair of dice? \_\_\_\_\_
- (64) 6 quarts = \_\_\_\_\_ pints
- (65)  $36^2 =$  \_\_\_\_\_
- (66)  $11^5 \div 9$  has remainder of \_\_\_\_\_
- (67) The area of a rhombus with diagonals 20 and  $15\frac{1}{2}$  is  
\_\_\_\_\_
- (68) If  $-3x + 15 < 12$ , then  $x >$  \_\_\_\_\_
- (69)  $(-24) \div (-0.25) =$  \_\_\_\_\_
- \*(70)  $21^3 =$  \_\_\_\_\_
- (71) The multiplicative inverse of  $-3\frac{5}{8}$  is \_\_\_\_\_
- (72)  $22^2 - 28^2 =$  \_\_\_\_\_
- (73) If 150% of 210 = \_\_\_\_\_
- (74) What is the area of a trapezoid with bases  $4\frac{3}{4}$ ,  $6\frac{3}{4}$  and altitude to the bases of 8? \_\_\_\_\_
- (75)  $21^2 + 42^2 =$  \_\_\_\_\_
- (76)  $(1 + 2 + 3 + \dots + 9)^2 =$  \_\_\_\_\_
- (77) What is the total surface area of a rectangular box with edges 3, 5 and 4? \_\_\_\_\_
- (78) The area of a right triangle with legs  $6\frac{2}{3}$  and 6 is  
\_\_\_\_\_
- (79)  $625 \times 48 =$  \_\_\_\_\_
- \*(80) 32 square miles = \_\_\_\_\_ acres

# 2024 – 2025 University Interscholastic League Elementary Number Sense Test C – Key

|                   |                      |                                   |                                   |
|-------------------|----------------------|-----------------------------------|-----------------------------------|
| (1) 97            | *(20) 26614 – 29414  | (38) 1.60                         | (59) $2\frac{5}{9}; \frac{23}{9}$ |
| (2) 121           | (21) 26              | (39) $8\frac{1}{3}; \frac{25}{3}$ | *(60) 604 – 666                   |
| (3) 655           | (22) 80              | *(40) 188813 – 208687             | (61) 1110                         |
| (4) 51            | (23) 35              | (41) $\frac{3}{4}$                | (62) 500                          |
| (5) 138           | (24) 2002            | (42) 220                          | (63) $\frac{1}{9}$                |
| (6) 109           | (25) $\frac{5}{12}$  | (43) 17                           | (64) 12                           |
| (7) 64            | (26) 10506           | (44) 20                           | (65) 1296                         |
| (8) 539           | (27) $\frac{21}{25}$ | (45) 125                          | (66) 5                            |
| (9) 3800          | (28) 36              | (46) $72\frac{7}{64}$             | (67) 155                          |
| *(10) 2651 – 2929 | (29) 80              | (47) 28                           | (68) 1                            |
| (11) 900          | *(30) 76570 – 84630  | (48) 20                           | (69) 96                           |
| (12) 39000        | (31) 111             | (49) 7                            | *(70) 8798 – 9724                 |
| (13) 3            | (32) 23              | *(50) 114194 – 126213             | (71) $-\frac{8}{29}$              |
| (14) 2499         | (33) 3               | (51) 14                           | (72) -300                         |
| (15) 2            | (34) 924             | (52) $13\frac{5}{8}$              | (73) 315                          |
| (16) 10           | (35) $\frac{5}{8}$   | (53) 3                            | (74) 46                           |
| (17) 300.082      | (36) 72              | (54) 120                          | (75) 2205                         |
| (18) 720          | (37) 675             | (55) 22                           | (76) 2025                         |
| (19) 44           |                      | (56) 3                            | (77) 94                           |
|                   |                      | (57) 50                           | (78) 20                           |
|                   |                      | (58) 2100                         | (79) 30000                        |
|                   |                      |                                   | *(80) 19456 – 21504               |

Note: \*(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like  $\frac{2}{3}$  it cannot be written as .666... or  $\overline{.6}$ .